

SKILLS

PROGRAMMING

Proficient:

MATLAB • QGIS • Google Earth Engine

Experienced:

Python • R • Stata • Tableau • $\LaTeX$

RESEARCH

Climate & Disaster Risk Assessment

Earthquake Risk & Loss Modelling

Deep Probabilistic Modelling

Geospatial Data Science

Spatiotemporal Analysis

OPEN-SOURCE PROJECTS

- **VBCI** [📄 ★ 14] [📄 66 cites] [📄 242]
Joint inference of multi-hazards & building damage from satellite images by causal dependency modelling (*Nature Communications*)
- **METEOR 2.5D: 72-GB Data** [📄 960]
Spatiotemporal maps of physical vulnerability in Least Developed Countries, 1975-2030 (AAAI'26)
- **DeepC4** [📄 2k]
Deep conditional census-constrained clustering for multi-task spatial disaggregation of urban morphology (AGU'24, ISPRS P&RS, *in review*)

EDUCATION

UNIVERSITY OF CAMBRIDGE

Ph.D. ARTIFICIAL INTELLIGENCE FOR

THE STUDY OF ENVIRONMENTAL RISKS

M.RES. ENVIRONMENTAL DATA SCIENCE

Oct. 2022 - Sep. 2026 | Cambridge, UK

UKRI EPSRC CDT Studentship (\$290k)

Dissertation: Understanding Regional Dynamic Physical Risks: Advanced Geospatiotemporal Modelling of Exposure & Vulnerability

STANFORD UNIVERSITY

M.A. PUBLIC POLICY AND

M.S. CIVIL & ENVIRONMENTAL ENG'G

EXECUTIVE EDUCATION CERTIFICATE

IN ENTREPRENEURSHIP & INNOVATION

Sep. 2019 - Jun. 2022 | Stanford, CA, USA

Knight-Hennessy Scholar (\$240k) (69/4,424)

UNIVERSITY OF THE PHILIPPINES

B.S. CIVIL ENGINEERING

Jun. 2013 - Jan. 2018 | Los Baños, PH

magna cum laude, college valedictorian (1/341)

completed one-semester early

TOKYO METROPOLITAN UNIVERSITY

EXCHANGE PROGRAM CERTIFICATE,

INTEGRATED RIVER ENGINEERING WORK

Oct. - Nov. 2017 | Tokyo, JP

SELECTED TALKS

Moody's Risk Management Solutions (RMS),
Global Earthquake Model (GEM) Conference,
RiskLayer, LandAware Innovations Group WG5

EXPERIENCE (8 1/2 years of research dev't, project mgm't, & risk consulting)

UNIVERSITY OF CAMBRIDGE | DOCTORAL RESEARCHER, GEOSPATIAL ML

Oct. 2022 - Sep. 2026 | Cambridge, UK | 4 years

- Improved the spatiotemporal resolution of physical exposure models of 44+ Least Developed Countries by 5x finer using geospatial ML on time-series satellite imagery.
- Established a university-industry collaboration with a nonprofit organization and a disaster risk agency in the Philippines for city-scale mapping of multi-hazard urban risk dynamics.

GERMAN AEROSPACE CENTER (DLR) | VISITING RESEARCHER

Jul. - Sep. 2024 | Apr. - Jun. 2026 | Munich, Germany | 6 months

- Led development of a geospatial ML framework & benchmark dataset for building vulnerability & change detection using satellite embeddings & census-based constraints.

EARTHQUAKES & MEGACITIES INITIATIVE (EMI) | URBAN RESILIENCE FELLOW

Mar. - Aug. 2022 | Quezon City, Philippines | 5 months

- Developed large-scale seismic risk assessment tools covering 400,000 buildings & 3.2M population for a scenario "Big One" M7.2 earthquake.

STANFORD LAND, BUILDINGS, & REAL ESTATE (LBRE) | SEISMIC RISK CONSULTANT

Mar. 2021 - Mar. 2022 | Stanford, CA, USA | 1 year

- Quantified earthquake-induced financial losses and business interruption risks for 750 buildings at Stanford University under seismic scenarios up to magnitude 7.6, and developed a Tableau dashboard for stakeholder decision-making.

FM GLOBAL, ENG'G & RESEARCH | STRUCTURES & NATURAL HAZARDS RESEARCHER

Jun. - Sep. 2021 | Norwood, MA, USA | 4 months

- Improved the firm's library of earthquake design hazard and maps by more than 246%, starting with 130 and ending with 320 maps (or 190 new maps) from over 130 countries.

STANFORD UNIVERSITY, STRUCTURES AS SENSORS RESEARCH LAB |

GEOSPATIAL & MACHINE LEARNING GRADUATE RESEARCHER

Jun. 2020 - Jun. 2021 | Stanford, CA, USA | 1 year

- Implemented a Bayesian causal inference framework leveraging satellite imagery to enhance post-earthquake landslide, liquefaction, & building damage models by 15x.

SOUTH BAY CONSULTING ENGINEERS | STRUCTURAL ENGINEERING INTERN

Jun. - Sep. 2020 | San Jose, CA, USA | 3 months

- Produced structural designs and resolved review comments while leading retrofit of a 9,000 ft² two-story home with a weak first story using a cantilevered column system.

NEW STORY CHARITY, RESEARCH & DEV'T TEAM | HOUSING RESEARCH FELLOW

Feb. - Jun. 2020 | San Francisco, CA, USA | 4 months

- Evaluated 7 affordable housing material innovations and led feasibility studies with 20 interdisciplinary fellows across finance, policy, architecture, engineering, and construction.

ARUP | STRUCTURAL ENGINEER

Jul. 2018 - Jul. 2019 | Metro Manila, Philippines | 1 year

- Conducted post-earthquake assessment, performance- and code-based structural peer reviews for multiple 41-50 story high-rise developments.

MAKATI DEVELOPMENT CORPORATION | PLANNING & CONTROL ENG'G INTERN

May. - Jul. 2017 | Bonifacio Global City, Philippines | 3 months

- Built a VBA-Excel system tracking deliverables across 100+ concurrent projects.

ENGINEERING CONGRESS | COMMITTEE HEAD

Jun. 2016 - Jun. 2017 | Los Baños, Philippines | 1 year

- Developed event invitation and registration strategies that increased participation fivefold (300 to 1,500 attendees) for major engineering seminars and quiz competitions.

PUBLICATIONS

Nature Communications, AAAI AI for Social Impact, Environmental Research Letters,
ISPRS Photogrammetry & Remote Sensing (*in review*), Progress in Disaster Science (*in review*)